

Course Handout

Institute/School	Chitkara University Institute of Engineering and Technology		
Department	Department of Computer Science & Engineering		
Programme	Bachelor of Engineering (B.E.) - Computer Science & Engineering		
Course Title	Introduction to Web Technologies	Session	2026-2027
Course Code	26CSE0106	Semester/Batch	1 st /2026
L-T-P(Per Week)/No. of Hours(Per Week)	2-0-4/8	Course Credits	04
Pre-requisite	NA	NHEQF Level	4.5
Course Coordinator	Dr. Chetna Sharma	SDG Number	4, 9

1. Scope and objectives of the Course

This course introduces learners to modern web technologies for designing, developing, testing, and deploying responsive, user-friendly web applications.

The main objectives of the course are:

- Understand Internet architecture, web protocols, DNS, and web hosting.
- Develop standards-compliant web pages using HTML5 and basic SEO.
- Design responsive and user-friendly interfaces using CSS3 and UI/UX principles.
- Build mobile-first web applications using the Bootstrap framework.
- Test, optimize, debug, and deploy websites using industry-standard tools.

2. Course Outcomes (COs)

After completing the course, student should be able to:

CO1: Explain the fundamental concepts of Web Technologies, Internet architecture, client-server architecture, web protocols, DNS, domain names, and web hosting.

CO2: Develop standards-compliant web pages using HTML5 by incorporating semantic elements, multimedia, forms, metadata, and basic SEO techniques.

CO3: Design user-friendly and visually appealing web interfaces by applying User Interface (UI) and User Experience (UX) principles and modern design practices.

CO4: Construct responsive web pages using CSS3 by applying selectors, layouts, Flexbox, media queries, animations, and responsive design principles.

CO5: Develop mobile-first responsive websites using the Bootstrap framework, grid system, reusable components, and utility classes.

CO6: Evaluate the quality and performance of websites by performing testing, debugging, optimization, validation, and deployment using browser developer tools and modern web development practices.

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	No. of Lectures	NHEQF Level Descriptor
CO1	H	M										H	L	9	Q1, Q2
CO2	M		H		M							H	M	19	Q2, Q3
CO3			H		M				M			M	M	10	Q3, Q4
CO4			H		M							H	H	21	Q3, Q4
CO5			H		M					M		H	H	11	Q3, Q4, Q5
CO6				H	M				L	M	M	M	H	20	Q3, Q4, Q5, Q6
Total Contact Hours														90	

H=High, M=Medium, L=Low

3. Recommended Books/E-Resources (Textbooks and Reference books):

B01: HTML and CSS: Design and Build Websites by Jon Duckett, 1st Edition, Wiley Publications.

B02: Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics by Jennifer Robbins, 5th Edition, O'Reilly Media.

B03: Responsive Web Design with HTML5 and CSS by Ben Frain, 4th Edition, Packt Publishing.

B04: Bootstrap 5: Beginner's Guide by Jacob Lett, BPB Publications.

B05: Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability by Steve Krug, 3rd Edition, New Riders.

B06: Web Technologies: A Computer Science Perspective by Jeffrey C. Jackson, Pearson Education.

E-Library Resource

<https://developer.mozilla.org/en-US/docs/Web>

<https://getbootstrap.com>

<https://validator.w3.org>

4. Recommended Tools and Platforms

Any Text Editor like Notepad++, Sublime, Visual Studio Code etc.

5. Delivery/Instructional Plan and Resources

Theory Plan:

Lect. No.	Topics	CO	Book No, CH No, Page No.	TLM	Web References/Audio-Video
1	Introduction to Web Technologies, Evolution of the World Wide Web, Internet Architecture, Client-Server Architecture, Frontend and Backend Technologies, Static and Dynamic Websites, Differences between Static and Dynamic Websites, Web Application Lifecycle, Stages of Web Application Development.	CO1	B05, CH 1- 2, Page No. 5-135	Lecture, Discussion	https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Web_standards/How_the_web_works https://onlinecourses.nptel.ac.in/noc22_ee61/preview https://onlinecourses.nptel.ac.in/noc25_cs15/preview
2	Web Browsers, Browser Rendering Process, Browser Developer Tools, HTTP and HTTPS Protocols, HTTP Methods (GET, POST, PUT, DELETE), HTTP Status Codes.	CO1	B05, CH 1- 2, Page No. 5-135	Lecture, Discussion, Questioning	https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Web_standards/How_the_web_works
3	HTTP Request-Response Model, HTTP Request and Response Components, DNS, Domain Names, Domain Name Resolution Process, Top-Level Domains (TLDs). URL Structure, Absolute and Relative URLs, Query Parameters, Web Hosting, Types of Web Hosting, Domain and Hosting Integration, Unit Summary.	CO1	B05, CH 1- 2, Page No. 5-135	Lecture, Discussion, Questioning	https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Web_standards/How_the_web_works https://onlinecourses.nptel.ac.in/noc25_cs15/preview
4-7	Introduction to HTML5, HTML5 Document Structure, HTML Elements and Attributes, Semantic HTML Elements, Headings, Paragraphs, Text Formatting, Comments, Hyperlinks, Images, Tables.	CO2	B01, CH 2-3, Page No. 12-37, CH 10, Page No. 160-190	Lecture, Discussion, Demonstration	https://onlinecourses.swayam2.ac.in/noc20_cs05/preview https://developer.mozilla.org/en-US/docs/Web/HTML https://onlinecourses.swayam2.ac.in/noc20_sp11/unit/unit=4&lesson=7
CAT 1					
8-10	Lists, Audio and Video Embedding, HTML Forms, Input Controls, Form Attributes, Form Validation, Meta Tags, Favicon, Search Engine Optimization (SEO) Basics, Best Practices for HTML5 Documents, Creating a Complete Responsive HTML5 Web Page.	CO2	B03 CH 1,2-3, Page No. 1-99, CH 6, Page No. 161-196, CH 12, Page No. 387-423	Lecture, Discussion, Demonstration	https://nptel.ac.in/courses/106106222 https://developer.mozilla.org/en-US/docs/Web/HTML https://onlinecourses.swayam2.ac.in/noc20_sp11/unit/unit=4&lesson=7
11	Introduction to User Interface (UI) and User Experience (UX), Need for Good UI and UX, User Engagement, User-Centered Design, Intuitiveness versus Efficiency.	CO3	B05 CH 1, Page No. 10-19, CH 2, Page No. 20-27, CH 3, Page No. 28-41	Lecture, Discussion, Demonstration	https://www.interaction-design.org/literature/topics/ux-design

12	Principles of User Interface Design, Design Goals and Constraints, Usability, Accessibility, Consistency, Feedback, Simplicity, Error Prevention.	CO3	B05 CH 1, Page No. 10–19, CH 3, Page No. 28–41, CH 5, Page No. 48–53, CH 9, Page No. 110–141, CH 12, Page No. 172–181	Lecture, Discussion, Demonstration	https://developer.apple.com/design/human-interface-guidelines/ https://developer.mozilla.org/en-US/docs/Web/Accessibility
13	Modern UI Design Styles, Minimalist Design, Flat Design, Navigation-Based Design, Best Practices for Designing User-Friendly Web Interfaces, Unit Summary.	CO3	B05 CH 3, Page No. 28–41, CH 5, Page No. 48–53, CH 6, Page No. 54–83, CH 7, Page No. 84–101	Lecture, Discussion, Demonstration	https://www.interaction-design.org/literature
14-15	Introduction to CSS3, CSS Syntax, Types of CSS (Inline, Internal, External), Selectors, Specificity, Units, Colors, Typography, CSS Gradients.	CO4	B02, CH 4-5, Page No. 429-533 B03, CH 14, Page No. 443-478	Lecture, Discussion, Demonstration	https://developer.mozilla.org/en-US/docs/Web https://css-tricks.com/
CAT2					
16-19	Backgrounds, Borders, Margins, Padding, Box Model, Display Properties, Positioning, Flexbox, Responsive Units, Viewport, Media Queries.	CO4	B04, CH 16, Page No. 443-470	Lecture, Discussion, Demonstration	https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Flexible_Box_Layout/Basic_Concepts_of_Flexbox
20-23	Transitions, Transforms, Animations, Pseudo-Classes, Pseudo-Elements, CSS Best Practices, Creating Responsive and Interactive Web Pages, Unit Summary.	CO4	B03, CH 14, Page No. 443-478	Lecture, Discussion, Demonstration	https://nptel.ac.in/courses/106106222 https://www.youtube.com/watch?v=JJS0Eo8JSnc
CAT3					
24-25	Introduction to the Bootstrap Framework, Features of Bootstrap, Containers, Grid System, Rows, Columns, Responsive Grid Layout, Mobile-First Approach.	CO5	B04, CH 16, Page No. 443-470	Lecture, Discussion, Demonstration	https://getbootstrap.com/docs/5.3/getting-started/introduction/ https://getbootstrap.com/docs/5.3/layout/grid/?utm_
26	Utility Classes, Typography, Navigation Bar, Cards, Buttons, Forms, Tables, Responsive Components.	CO5	B04, CH 16, Page No. 443-470	Lecture, Discussion, Demonstration, Case-study	https://getbootstrap.com/docs/5.3/components/accordion/ https://www.freecodecamp.org/learn/
27	Modals, Bootstrap Customization, Mobile-First Design Principles, Best Practices for Developing Responsive Websites using Bootstrap, Unit Summary.	CO5	B04, CH 16, Page No. 443-470	Lecture, Discussion, Demonstration	https://getbootstrap.com/docs/5.3/components/modal/ https://www.youtube.com/@getbootstrap
28	Introduction to Website Testing, Cross-Browser Compatibility Testing, Browser Developer Tools, Debugging Techniques, Website Validation.	CO6	B06, CH5, Page No. 129- 165	Lecture, Discussion, Demonstration	https://developer.mozilla.org/en-US/docs/Learn_web_development?utm_
29	Performance Optimization, Image Optimization, Asset Optimization (CSS, JavaScript and Media Files), Best Practices for Improving Website Performance.	CO6	B06, CH5, Page No. 129- 165	Lecture, Discussion, Demonstration	https://www.freecodecamp.org/learn https://web.dev/learn/performance/?utm_

Course Plan

30	Static Website Deployment, Website Maintenance, Deployment Best Practices, Website Performance Review, Course Summary and Revision.	CO6	B06, CH5, Page No. 129- 165	Lecture, Discussion, Demonstration	https://git-scm.com/book/en/v2 https://docs.github.com/en/pages
CAT4, CAT5					

Lab Plan:

Lab No.	Experiment	CO	TLM	Web References/Audio-Video
1-2	Explain Client-Server Architecture and observe HTTP Requests and Responses using Browser Developer Tools.	CO1	Discussion	https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Web_standards/How_the_web_works
3-4	Create a basic HTML5 webpage for a personal profile demonstrating the HTML5 document structure, headings, paragraphs, text formatting, comments, and hyperlinks.	CO2	Demonstration method using a simulation/tool	https://www.coursera.org/learn/html-css-javascript-for-web-developers
5-6	Build a department webpage using semantic HTML5 elements such as <header>, <nav>, <section>, <article>, <aside>, and <footer>.	CO2	Demonstration method using a simulation/tool	https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/link
7-8	Develop a course information webpage using ordered and unordered lists, nested lists, tables, hyperlinks, and navigation links.	CO2	Demonstration method using a simulation/tool	https://developer.mozilla.org/en-US/docs/Glossary/Semantics
9-10	Create a college event webpage incorporating images, audio, and video using appropriate HTML5 elements and attributes.	CO2	Demonstration method using a simulation/tool	https://onlinecourses.nptel.ac.in/noc22_ee61/preview https://onlinecourses.nptel.ac.in/noc25_cs15/preview
11-12	A college wants to collect student information through an online registration portal. Create an HTML5 form using text fields, password fields, radio buttons, checkboxes, dropdown lists, text areas, and buttons.	CO2	Demonstration method using a simulation/tool	https://www.coursera.org/learn/html-css-javascript-for-web-developers
13-14	Enhance the student registration form by implementing HTML5 validation using required, pattern, min, max, minlength, maxlength, email, and other appropriate attributes.	CO2	Demonstration method using a simulation/tool	https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/table
15-16	Prepare an online college course webpage using suitable meta tags, favicon, page title, semantic elements, headings, and basic SEO practices.	CO2	Demonstration method using a simulation/tool	https://www.coursera.org/learn/duke-programming-web
17-18	Create a college/student portal webpage with a well-organized navigation menu, clearly structured content sections, buttons, forms, and feedback messages. Apply basic UI/UX principles to ensure that the interface is intuitive, consistent, and easy to navigate.	CO3	Demonstration method using a simulation/tool	https://developer.apple.com/design/human-interface-guidelines/ https://developer.mozilla.org/en-US/docs/Web/Accessibility
19-20	Improve the college/student webpage by implementing semantic HTML elements, meaningful labels, alternative text for images, accessible form controls, keyboard-friendly navigation, and appropriate accessibility features to make the webpage usable by a wider range of users.	CO3	Demonstration method using a simulation/tool	https://developer.apple.com/design/human-interface-guidelines/ https://developer.mozilla.org/en-US/docs/Web/Accessibility
21-22	Create a webpage and style it using inline, internal, and external CSS, demonstrating selectors, colors, units, typography, and basic text styling.	CO4	Demonstration method using a simulation/tool	https://www.coursera.org/learn/html-and-css-in-depth

23-24	Create a student profile webpage demonstrating element, class, ID, attribute, and combined selectors along with specificity, and inheritance.	CO4	Demonstration method using a simulation/tool	https://nptel.ac.in/courses/106106222
25-26	Design a product/service card layout using margins, padding, borders, backgrounds, width, height, and the CSS box-sizing property.	CO4	Demonstration method using a simulation/tool	https://www.coursera.org/learn/html-and-css-in-depth
27-28	Develop a responsive food menu or product catalogue using CSS Flexbox for alignment, spacing, direction, wrapping, and sizing.	CO4	Demonstration method using a simulation/tool	https://www.coursera.org/learn/duke-programming-web
29-30	A news portal needs to work across mobile phones, tablets, and desktops. Create a responsive webpage using viewport units, relative units, breakpoints, and media queries.	CO4	Demonstration method using a simulation/tool	https://www.coursera.org/learn/html-and-css-in-depth
31-32	Develop any webpage with pseudo-classes and pseudo-elements. Use pseudo-elements ::before and ::after to insert decorative or functional content without modifying HTML.	CO4	Demonstration method using a simulation/tool	https://www.coursera.org/learn/duke-programming-web
33-34	Develop a website by applying CSS transforms, transitions, and keyframe animations to images, buttons, cards, and other interface elements.	CO4	Demonstration method using a simulation/tool	https://www.coursera.org/learn/html-and-css-in-depth
35-36	Develop a Web Page Layout using Bootstrap Grid System and Utility Classes	CO5	Demonstration method using a simulation/tool	https://getbootstrap.com/docs/5.3/components/accordion/
37-38	Build a student portal interface using Bootstrap Navbar, responsive navigation, forms, buttons, cards, and grid components.	CO5	Demonstration method using a simulation/tool	https://getbootstrap.com/docs/5.3/components/accordion/
39-40	A college website has been developed using HTML5 and CSS3, but some elements are incorrectly aligned, links are not working, and the webpage contains HTML and CSS errors. Identify and correct the errors using Browser Developer Tools and validate the corrected HTML and CSS code.	CO6	Demonstration method using a simulation/tool	https://developer.mozilla.org/en-US/docs/Learn_web_development?utm_
41-44	A responsive college event webpage contains multiple images, CSS files, and media resources. Test the webpage on different browsers and screen sizes, identify compatibility issues, and optimize its images, CSS, and media resources to improve loading performance.	CO6	Demonstration method using a simulation/tool	https://www.freecodecamp.org/learn https://web.dev/learn/performance/?utm_
45-48	Deploy a tested and optimized static website on GitHub Pages. Create and configure the repository, upload the required website files.	CO6	Demonstration method using a simulation/tool	https://developer.mozilla.org/en-US/docs/Learn_web_development?utm_
49-52	Perform maintenance of the deployed website by updating webpage content, checking and correcting broken links, validating HTML and CSS, and verifying accessibility and responsive behavior across different screen sizes.	CO6	Demonstration method using a simulation/tool	https://www.freecodecamp.org/learn https://web.dev/learn/performance/?utm_
53-60	Develop and deploy a complete responsive website integrating HTML5, CSS3, Bootstrap, testing, optimization, and deployment techniques as a mini project.	CO1 - CO6	Demonstration method using a simulation/tool	https://git-scm.com/book/en/v2 https://docs.github.com/en/pages

6. Course ERISE Mapping: Employability

7. Active Learning Methodology Used:

- **Think-Pair-Share:** Students discuss web technologies, UI/UX principles and responsive design concepts in pairs and present their understanding.
- **Hands-on Learning:** Live coding sessions on HTML5, CSS3 and Bootstrap followed by individual and group practice.

- **Case Study & Problem-Based Learning:** Analysis of popular websites to evaluate usability, responsiveness and performance.
- **Mini Project:** Students design, develop, test and deploy a responsive website integrating all course concepts.

8. Action plan for different types of learners

Fast Learners	Average Learners	Learners with constant monitoring
• Projects • Encouragement for improvement using Peer Tutoring	• Workshops • Doubt Sessions • E-notes and E-exercises to read ahead of the pedagogic material.	• Remedial Classes • Doubt Sessions • Guided Tutorials

9. Additional Self-directed Learning

Assignments to promote self-learning, survey of contents from multiple sources.

S. No	Advanced Topics, Additional Reading, Research papers and any	CO	References/MOOCs
1	Introduction to Javascript	CO5, CO6	https://nptel.ac.in/courses/106105084 https://www.youtube.com/watch?v=qoSksQ4s_hg

10. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	No. of Assessments	Type of Component	Component Description	% Weightage of Component	Max. Marks	Mode of Assessment	CO
Formative	Component1	01	Continuous Assessment(CE)	CAT1	25%	25	Computer Based Test	CO1-CO2
Formative	Component2	01	Continuous Assessment(CE)	CAT2	25%	25	Computer Based Test	CO2, CO3, CO4
Formative	Component3	01	Continuous Assessment(CE)	CAT3	25%	25	Computer Based Test	CO3, CO4
Formative	Component4	01	Continuous Assessment(CE)	CAT4	25%	25	Computer Based Test	CO1-CO6
Formative	Component5	01	Continuous Assessment(CE)	CAT5	25%	25	Computer Based Test	CO1-CO6
		Total		100%				

Note:

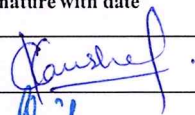
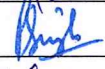
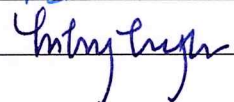
- The academic performance for each subject is evaluated across five Continuous Assessments (CA1 through CA5).
- **Best-of-Four Policy:** The final score will be calculated based on the best four scores out of the five assessments.
- **Mandatory Requirement:** Students must appear in at least one of CA4 or CA5. Failure to appear in both CA4 and CA5 will result in an Incomplete (I) grade being awarded for the respective subject.
- **Passing Threshold:** Students must secure a minimum aggregate of 50% marks to pass a subject. Scoring below 50% will result in a Fail grade ('F').

11. Sessional Test-Wise Syllabus of the Course:

Evaluation Component	Description	Syllabus Covered	Weightage %
Component1	CAT1	Lecture (1 - 7)	25%
Component2	CAT2	Lecture (8 - 15)	25%
Component3	CAT3	Lecture (16 - 23)	25%
Component4	CAT4	Lecture (1 - 30)	25%

Component5	CAT5	Lecture (1 - 30)	25%
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This Document is approved by:

Designation	Name	Signature with date
Prepared by Course Coordinator	Dr. Chetna Sharma	
Verified by Program Incharge	Dr. Preetinder Singh Brar	
Approved by Pro Vice Chancellor	Dr. Jaiteg Singh	
Date (DD/MM/YYYY)	18/08/2026	